Reasoning LLMs

Module 1 — What are they, why now?

Lucas Soares · O'Reilly Live Training

What is "reasoning"?

> Multi-step problem solving — not retrieval.

Vanilla LLM weak spots

- Knowledge cutoffs
- No actions in the world
- Long-horizon planning collapses

Direct answer vs. thinking chain

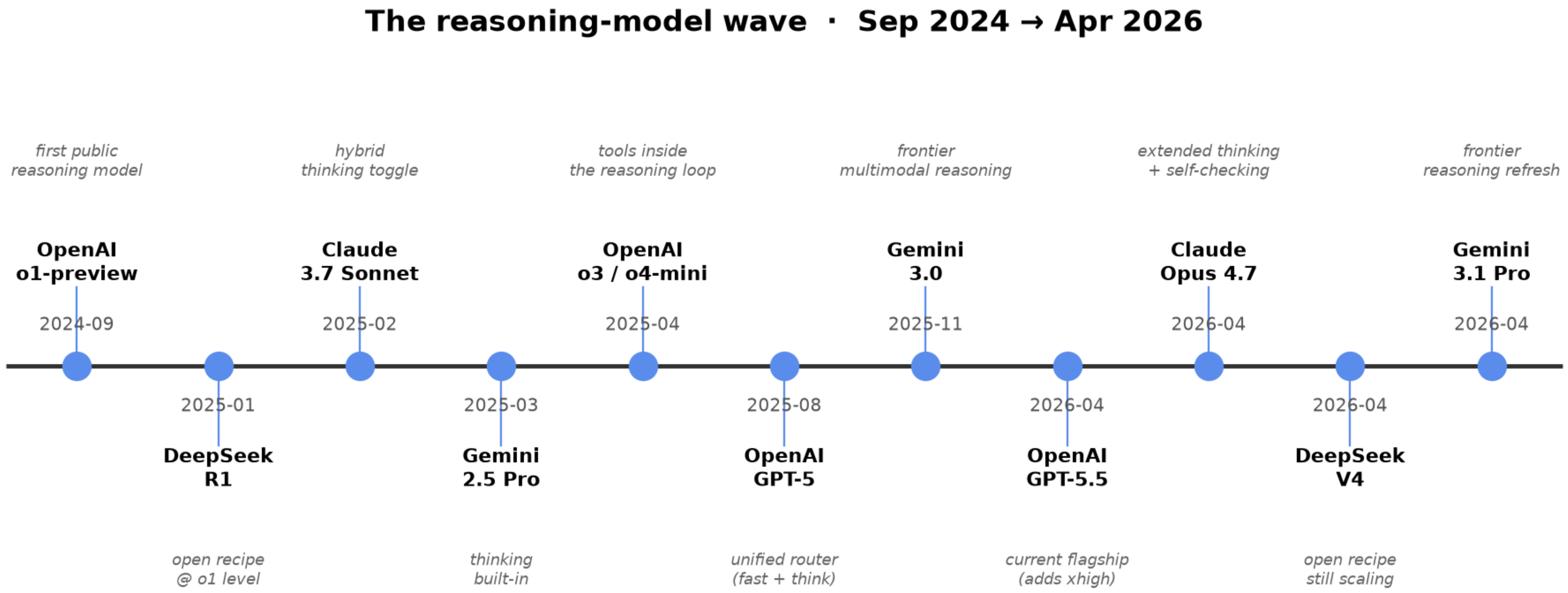
gpt-5.5 (low effort) — direct answer

```
Q: A train leaves Lisbon at 3pm
   going 80 km/h. Another leaves
   Porto at 4pm going 100 km/h.
   When do they meet?

A: They meet at 5:30pm.
```

gpt-5.5 (high effort) / R1 — with <think> trace

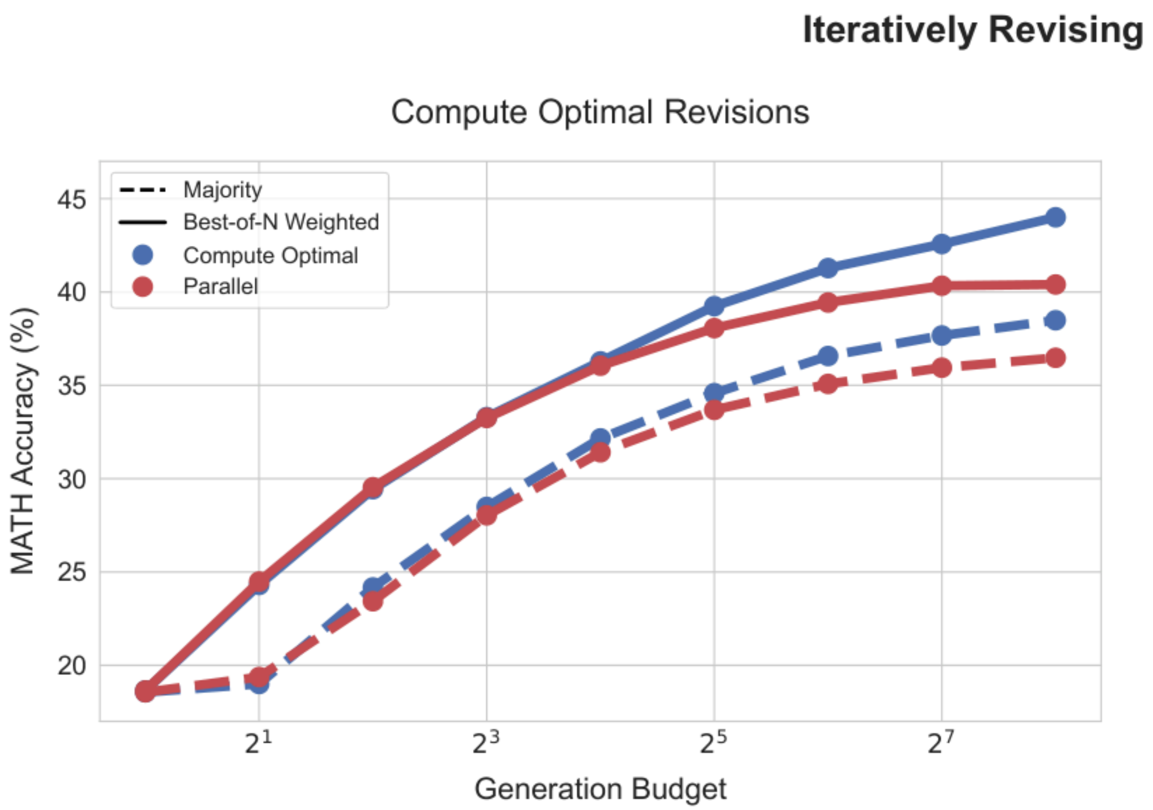
```
<think>
 Distance Lisbon↔Porto ≈ 300 km.
 At 3pm train A starts. By 4pm
 it has gone 80 km, so 220 km
 remain. Closing speed 180 km/h.
 t = 220/180 ≈ 1.22 h ≈ 1h13m.
 4pm + 1h13m = 5:13pm.
</think>

A: They meet around 5:13pm.
```

Timeline



The reasoning-model wave · Sep 2024 → Apr 2026
first public reasoning model
OpenAI o1-preview
2024-09
2025-01
DeepSeek R1
open recipe @ o1 level
hybrid thinking toggle
Claude 3.7 Sonnet
2025-02
2025-03
Gemini 2.5 Pro
thinking built-in
tools inside the reasoning loop
OpenAI o3 / o4-mini
2025-04
2025-08
OpenAI GPT-5
unified router (fast + think)
frontier multimodal reasoning
Gemini 3.0
2025-11
2026-04
OpenAI GPT-5.5
current flagship (adds xhigh)
extended thinking + self-checking
Claude Opus 4.7
2026-04
2026-04
DeepSeek V4
open recipe still scaling
frontier reasoning refresh
Gemini 3.1 Pro
2026-04

Test-time compute scaling

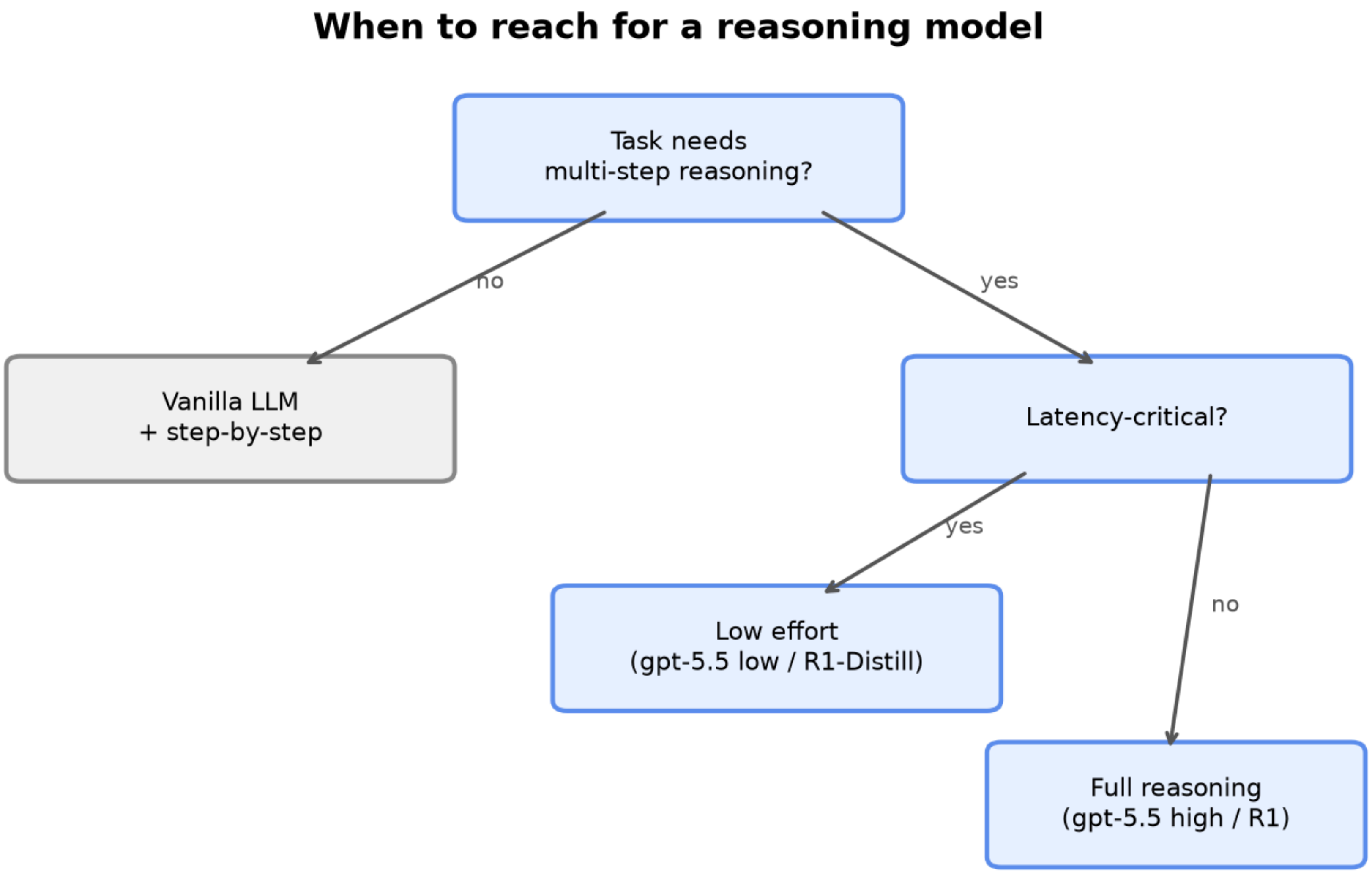


> More thinking → more accuracy.

The simplest trick

```
Q: A train leaves at 3pm…
A: Let's think step by step.
```

When to reach for a reasoning model



Cost & latency

> 5–50× more tokens. Use only when the task needs it.

Reasoning effort knobs

- **OpenAI:** `reasoning_effort = none / low / medium / high / xhigh`
- **Anthropic:** `thinking.budget_tokens = N`

Live demo →

`notebooks/01_chain_of_thought_intuition.ipynb`

GPT-2 with and without "Let's think step by step"

Recap

- Reasoning ≠ retrieval
- Trade tokens for accuracy
- Next: how DeepSeek **trained** a reasoning model